AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Forecasting with regression

Representing temporal structure with regressors

Session 5 of 10



Outline

- The linear model with time series
- Evaluating the regression model
- Selecting predictors
- Forecasting with regression
- Correlation, causation and forecasting
- Some useful predictors for regression models

SECTION 1 OF 6

The linear model with time series

1 **The linear model with time series**
2 Evaluating the regression model
3 Selecting predictors
4 Forecasting with regression
5 Correlation, causation and forecasting
6 Some useful predictors for regression models



Regression models

- To **explain**
- To **forecast**
- Simple linear regression model(SLR)
- Multiple linear regression model (MLR)

SLR model in theory

Regression model allows for a linear relationship between the forecast variable y and a single predictor variable x.

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t.$$

- y_t is the variable we want to predict: the “response” variable
- Each x_t is numerical and is called a “predictor”
- β_0 and β_1 are regression coefficients

SLR model in practice

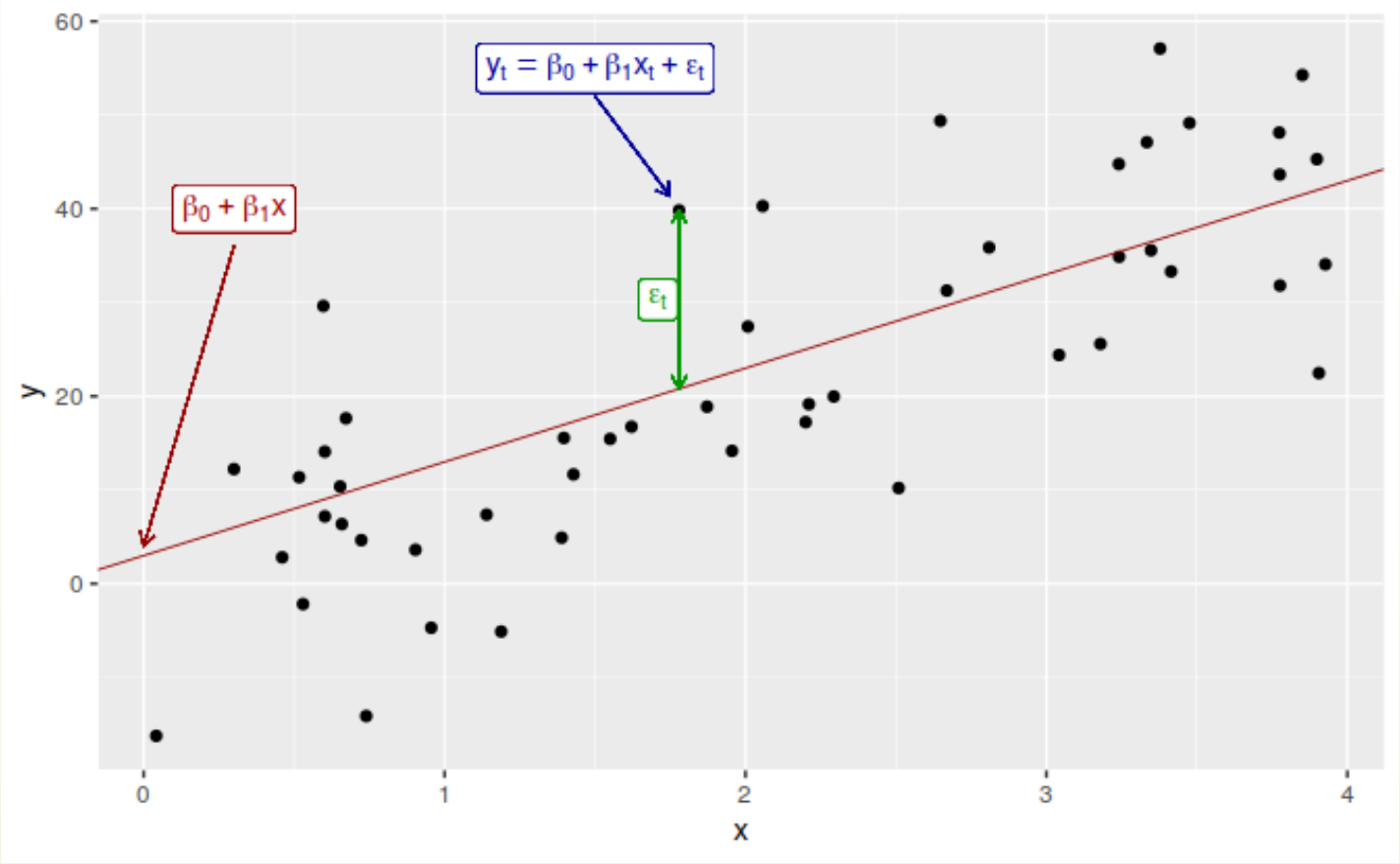
In practice, of course, we have a collection of observations but we do not know the values of the coefficients $\hat{\beta}_0$, $\hat{\beta}_1$. These need to be estimated from the data.

$$y_t = \hat{\beta}_0 + \hat{\beta}_1 x_t.$$

- y_t is the "response" variable
- Each x_t is a "predictor"
- $\hat{\beta}_0$ is the estimated intercept
- $\hat{\beta}_1$ is the estimated slope

What is the best fit

- There are many ways that a straight line can be laid on the scatter
- Best known criterion is called Ordinary Least Squares(OLS)



Estimation of the model

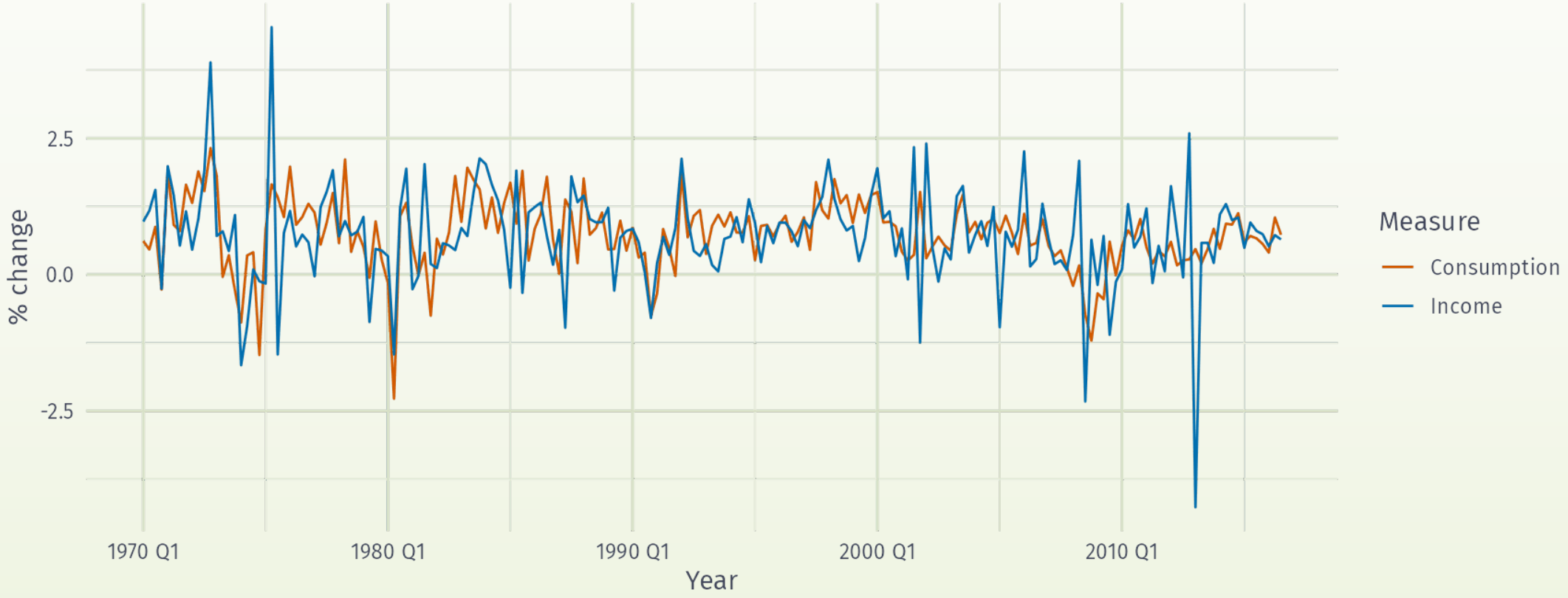
That is, we find the values of β_0 and β_1 which minimise

$$\sum_{i=1}^{N} e_i^2 = \sum_{i=1}^{N} (y_i - \beta_0 - \beta_1 x_i)^2.$$

- This is called *least squares* estimation because it gives the least value of the sum of squared errors.
- Finding the best estimates of the coefficients is often called *fitting* the model to the data.
- We refer to the *estimated* coefficients using the notation $\hat{\beta}_0, \hat{\beta}_1$.

Example: US consumption expenditure

```
us_change %>%
  gather("Measure", "Change", Consumption, Income) %>%
  autoplot(Change) +
  ylab("% change") + xlab("Year")
```



Example: US consumption expenditure



Example: US consumption expenditure

```
fit_cons <- us_change %>%
  model(lm = TSLM(Consumption ~ Income))
report(fit_cons)
```

```
Series: Consumption
Model: TSLM

Residuals:
   Min     1Q Median     3Q    Max
-2.408 -0.318  0.026  0.300  1.452

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.5451     0.0557    9.79  < 2e-16 ***
Income        0.2806     0.0474    5.91  1.6e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.603 on 185 degrees of freedom
Multiple R-squared: 0.159,  Adjusted R-squared: 0.154
F-statistic:   35 on 1 and 185 DF, p-value: 2e-08
```

Multiple regression

- In multiple regression there is one variable to be forecast and several predictor variables.
- The basic concept is that we forecast the time series of interest y assuming that it has a linear relationship with other time series x_1, x_2, ..., x_K
- We might forecast daily A&E attendance y using temperature x_1 and GP visits x_2 as predictors.

How many variable can we add?

You can add as many as you want but be aware of:

- Over-fitting
- Multicollinearity

Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

- y_t is the variable we want to predict: the "response" variable
- Each $x_{j,t}$ is numerical and is called a "predictor". They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \ldots, \beta_k$ measure the effect of each predictor after taking account of the effect of all other predictors in the model.

That is, the coefficients measure the **marginal effects**.

- ε_t is a white noise error term

Estimation of the model

We find the values of $\hat{\beta}_0, \ldots, \hat{\beta}_k$ which minimise

$$\sum_{i=1}^{N} e_i^2 = \sum_{i=1}^{N} (y_i - \beta_0 - \beta_1 x_{1,i} - \cdots - \beta_k x_{k,i})^2.$$

- This is called *least squares* estimation because it gives the least value of the sum of squared errors
- Finding the best estimates of the coefficients is often called *fitting* the model to the data
- We refer to the *estimated* coefficients using the notation $\hat{\beta}_0, \ldots, \hat{\beta}_k$.

Useful predictors in linear regression

Linear trend

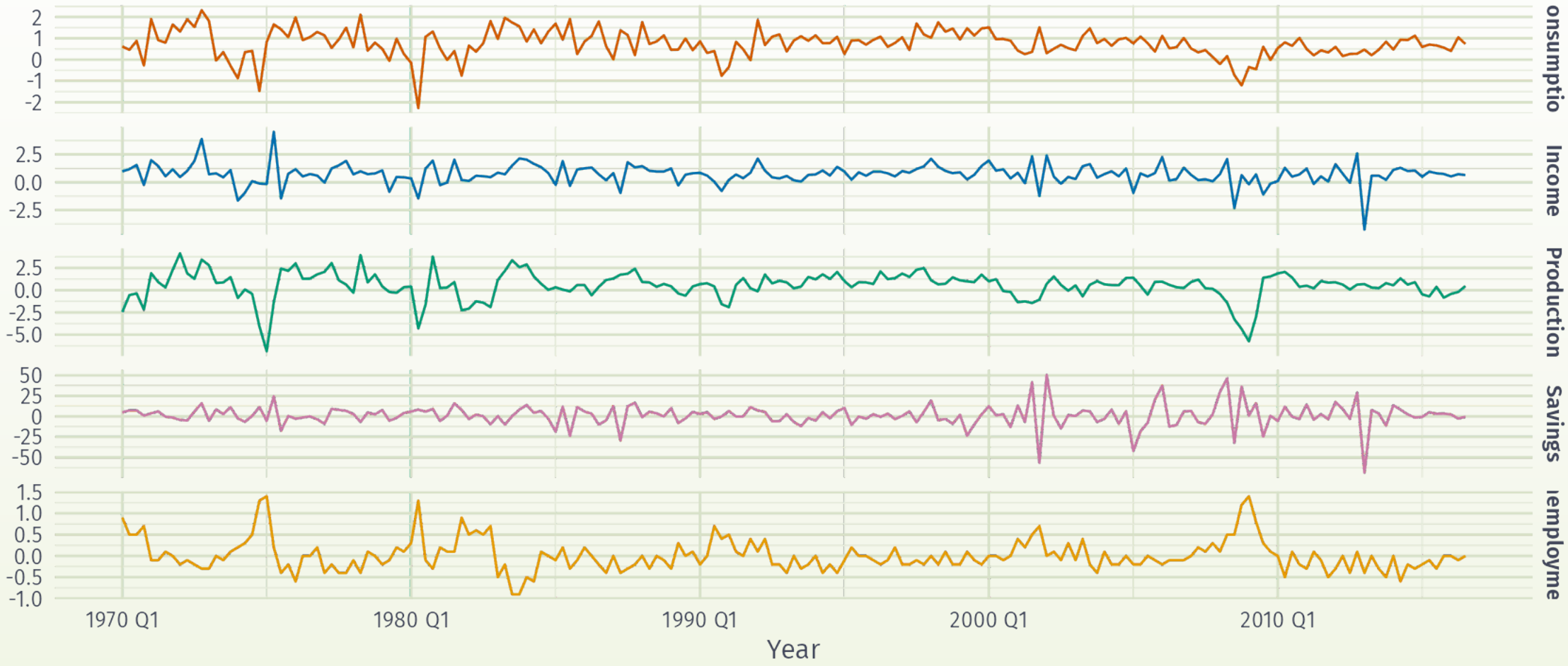
$$x_t = t$$

- $t = 1, 2, \ldots, T$
- Strong assumption that trend will continue.
- use special function `trend()`

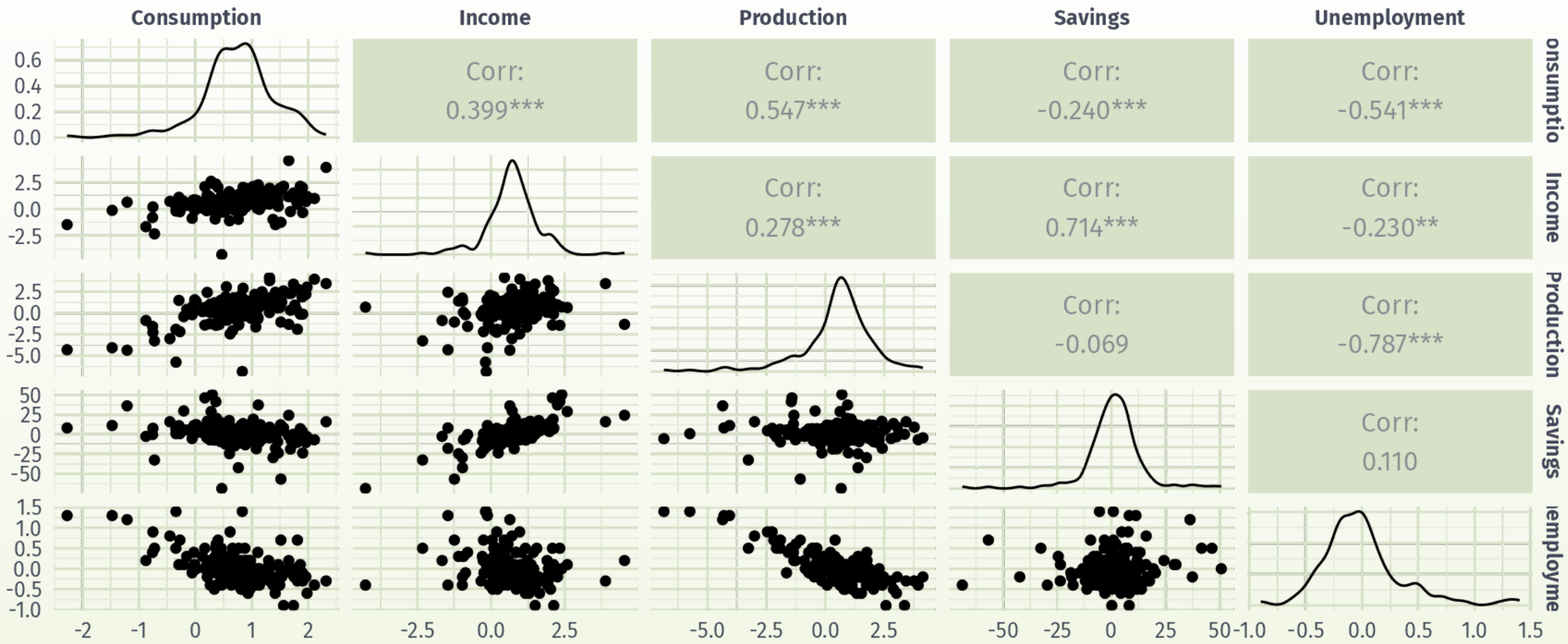
Seasonality

- Seasonality will be considered based on the interval of index
- use special function `season()`

Example: US consumption expenditure



Example: US consumption expenditure



Example: US consumption expenditure

```
fit_consMR <- us_change %>%
  model(lm = TSLM(Consumption ~ Income + Production + Unemployment + Savings))
report(fit_consMR)
```

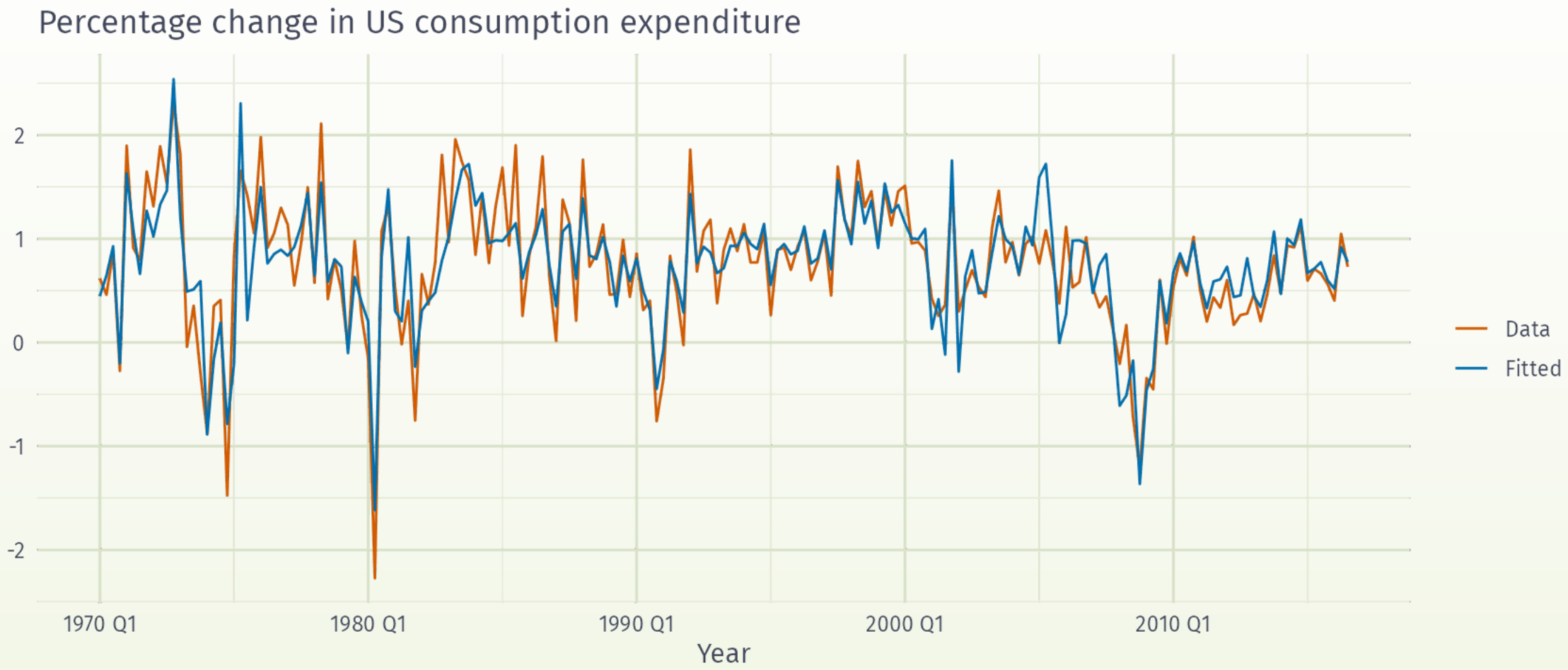
```
Series: Consumption
Model: TSLM

Residuals:
   Min     1Q Median     3Q    Max
-0.883 -0.176 -0.037  0.153  1.206

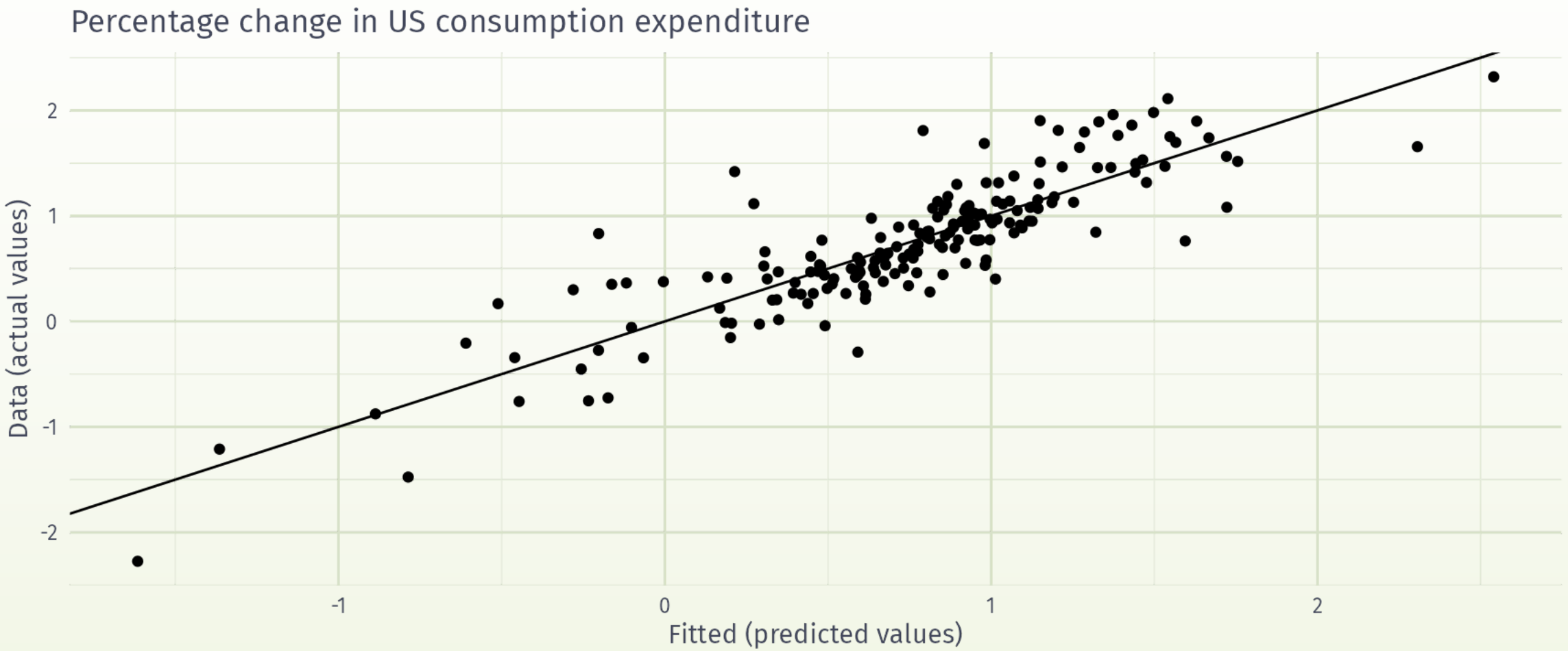
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.26729    0.03721    7.18  1.7e-11 ***
Income        0.71448    0.04219   16.93  < 2e-16 ***
Production    0.04589    0.02588    1.77    0.078 .
Unemployment -0.20477    0.10550   -1.94    0.054 .
Savings      -0.04527    0.00278  -16.29  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.329 on 182 degrees of freedom
Multiple R-squared: 0.754,  Adjusted R-squared: 0.749
F-statistic:  139 on 4 and 182 DF, p-value: <2e-16
```

Example: US consumption expenditure

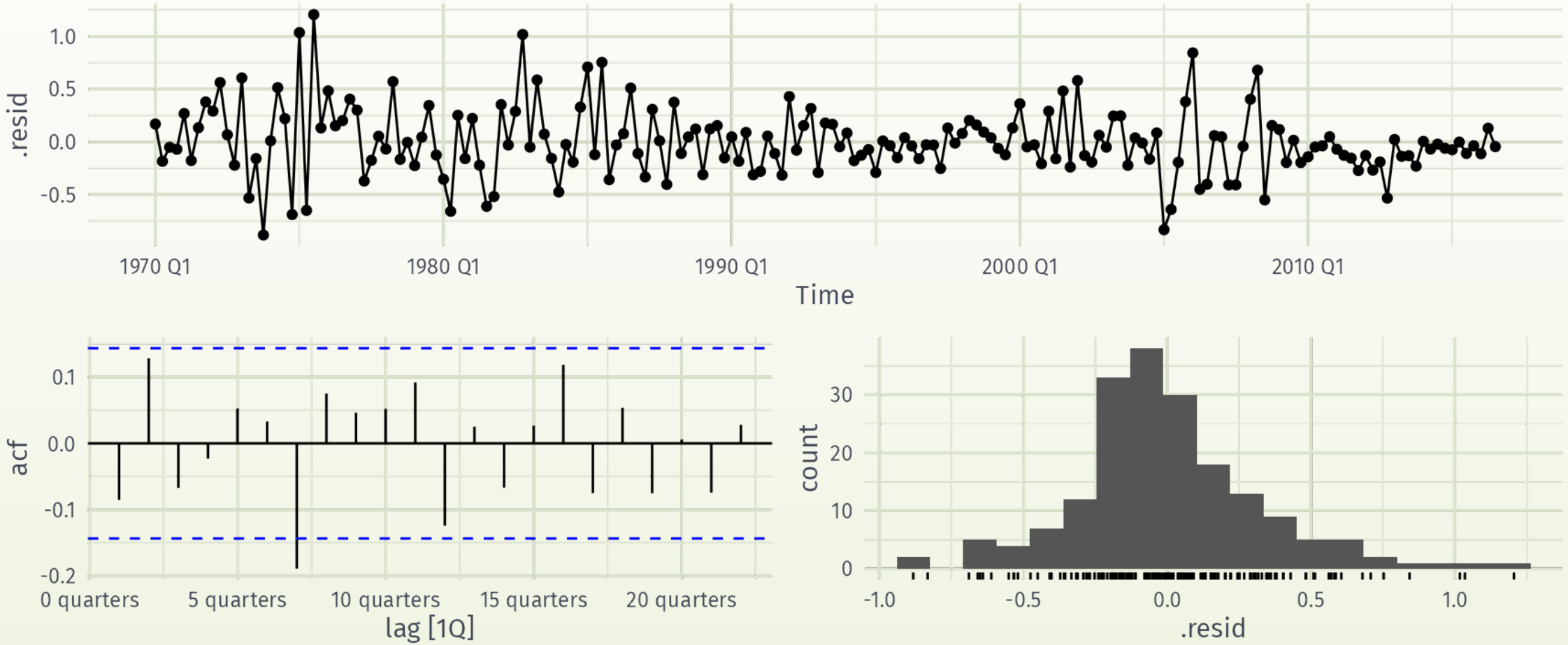


Example: US consumption expenditure



Example: US consumption expenditure

```
augment(fit_consMR) %>%
  gg_tsdisplay(.resid, plot_type="hist")
```



SECTION 2 OF 6

Evaluating the regression model

1 The linear model with time series
2 **Evaluating the regression model**
3 Selecting predictors
4 Forecasting with regression
5 Correlation, causation and forecasting
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Multiple regression and forecasting

For forecasting purposes, we require the following assumptions:

- ε_t are uncorrelated and zero mean
- ε_t are uncorrelated with each $x_{j,t}$.

It is **useful** to also have $\varepsilon_t \sim \mathrm{N}(0, \sigma^2)$ when producing prediction intervals or doing statistical tests.

Residual diagnostics

There are a series of plots that should be produced in order to check different aspects of the fitted model and the underlying assumptions.

1. check if residuals are uncorrelated using `ACF`
2. Check if residuals are normally distributed

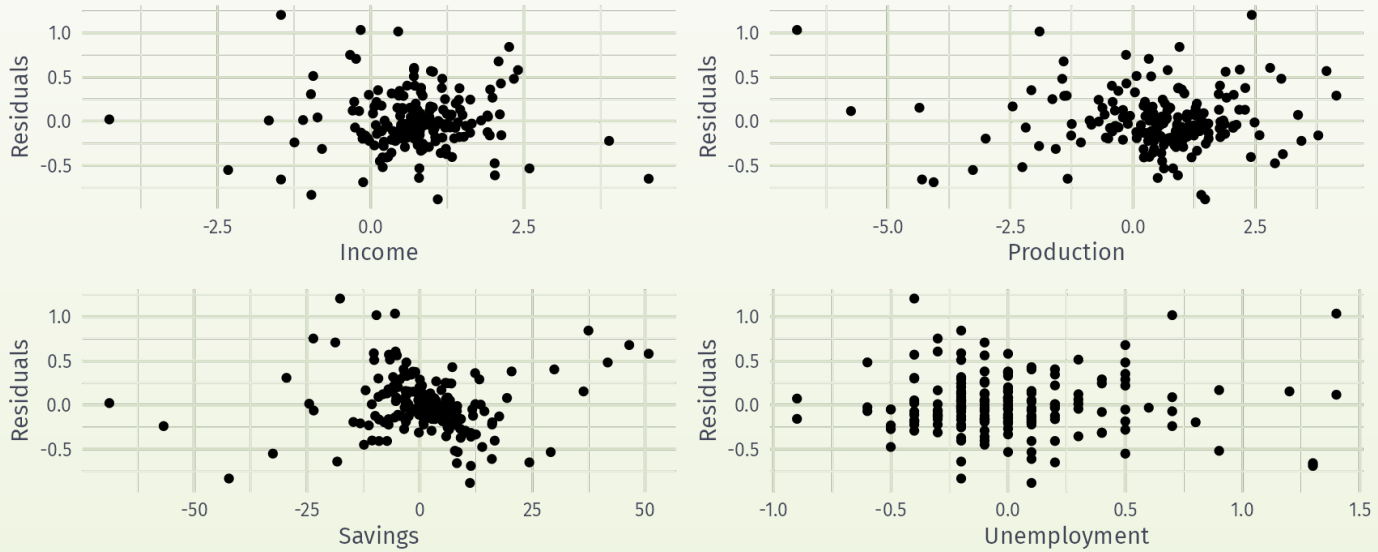
Residual scatterplots

Useful for spotting outliers and whether the linear model was appropriate.

- Scatterplot of residuals ε_t against each predictor $x_{j,t}$.
- Scatterplot residuals against the fitted values $\hat{y}_t$
- Expect to see scatterplots resembling a horizontal band with no values too far from the band and no patterns such as curvature or increasing spread.

Example: US consumption expenditure

```
df <- left_join(us_change, residuals(fit_consMR), by = "Time")
p1 <- ggplot(df, aes(x=Income, y=.resid)) +
  geom_point() + ylab("Residuals")
p2 <- ggplot(df, aes(x=Production, y=.resid)) +
  geom_point() + ylab("Residuals")
p3 <- ggplot(df, aes(x=Savings, y=.resid)) +
  geom_point() + ylab("Residuals")
p4 <- ggplot(df, aes(x=Unemployment, y=.resid)) +
  geom_point() + ylab("Residuals")
(p1 | p2) / (p3 | p4)
```



Residual patterns

- If a plot of the residuals vs any predictor in the model shows a pattern, then the relationship is non-linear.
- If a plot of the residuals vs any predictor **not** in the model shows a pattern, then the predictor should be added to the model.
- If a plot of the residuals vs fitted values shows a pattern, then there is heteroscedasticity in the errors. (Could try a transformation.)

SECTION 3 OF 6

Selecting predictors

1 The linear model with time series
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Comparing regression models

Computer output for regression will always give the R^2 value. This is a useful summary of the model.

- It is equal to the square of the correlation between y and $\hat{y}$.
- It is often called the "coefficient of determination".
- It can also be calculated as follows: $R^2 = \frac{\sum(\hat{y}_t - \bar{y})^2}{\sum(y_t - \bar{y})^2}$
- It is the proportion of variance accounted for (explained) by the predictors.

Comparing regression models

However ...

- R^2 does not allow for `degrees of freedom`.
- Adding *any* variable tends to increase the value of R^2, even if that variable is irrelevant.

To overcome this problem, we can use *adjusted* R^2:

$$\bar{R}^2 = 1 - (1 - R^2)\frac{T-1}{T-k-1}$$

where $k =$ no. predictors and $T =$ no. observations.

Maximising $\bar{R}^2$ is equivalent to minimising $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T-k-1}\sum_{t=1}^{T}\varepsilon_t^2$$

Cross-validation

1. Remove observation t from the data set, and fit the model using the remaining data. Then compute the error for the omitted observation
2. Repeat step 1 for $t = 1, \ldots, T$
3. Compute the MSE from errors obtained in 1. We shall call this the CV

Akaike's Information Criterion

$$\text{AIC} = -2\log(L) + 2(k+2)$$

where L is the likelihood and k is the number of predictors in the model.

- This is a *penalised likelihood* approach.
- *Minimising* the AIC gives the best model for prediction.
- AIC penalises terms more heavily than $\bar{R}^2$.
- Minimising the AIC is asymptotically equivalent to minimising MSE via leave-one-out cross-validation.

Corrected AIC

For small values of T, the AIC tends to select too many predictors, and so a bias-corrected version of the AIC has been developed.

$$\text{AIC}_\text{C} = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

As with the AIC, the AIC_C should be minimised.

Comparing regression models

```
glance(fit_consMR) %>%
  select(r_squared, adj_r_squared, AIC, AICc, CV)
```

```
# A tibble: 1 × 5
  r_squared adj_r_squared   AIC  AICc    CV
      <dbl>         <dbl> <dbl> <dbl> <dbl>
1     0.754         0.749 -409. -409. 0.116
```

Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

Choosing regression variables

Backwards stepwise regression

- Start with a model containing all variables.
- Try subtracting one variable at a time. Keep the model if it has lower CV or AICc.
- Iterate until no further improvement.
- You can also do forward stepwise

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Forecasting with regression

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Ex-ante versus ex-post forecasts

- *Ex ante forecasts* are made using only information available in advance.
  - require forecasts of predictors
- *Ex post forecasts* are made using later information on the predictors.
  - useful for studying behaviour of forecasting models.
- trend, seasonal and calendar variables are all known in advance, so these don't need to be forecast.

Scenario based forecasting

- Assumes possible scenarios for the predictor variables
- Prediction intervals for scenario based forecasts do not include the uncertainty associated with the future values of the predictor variables.

Building a predictive regression model

- If getting forecasts of predictors is difficult, you can use lagged predictors instead.

$$y_t = \beta_0 + \beta_1 x_{1,t-h} + \cdots + \beta_k x_{k,t-h} + \varepsilon_t.$$

- A different model for each forecast horizon h.

US Consumption

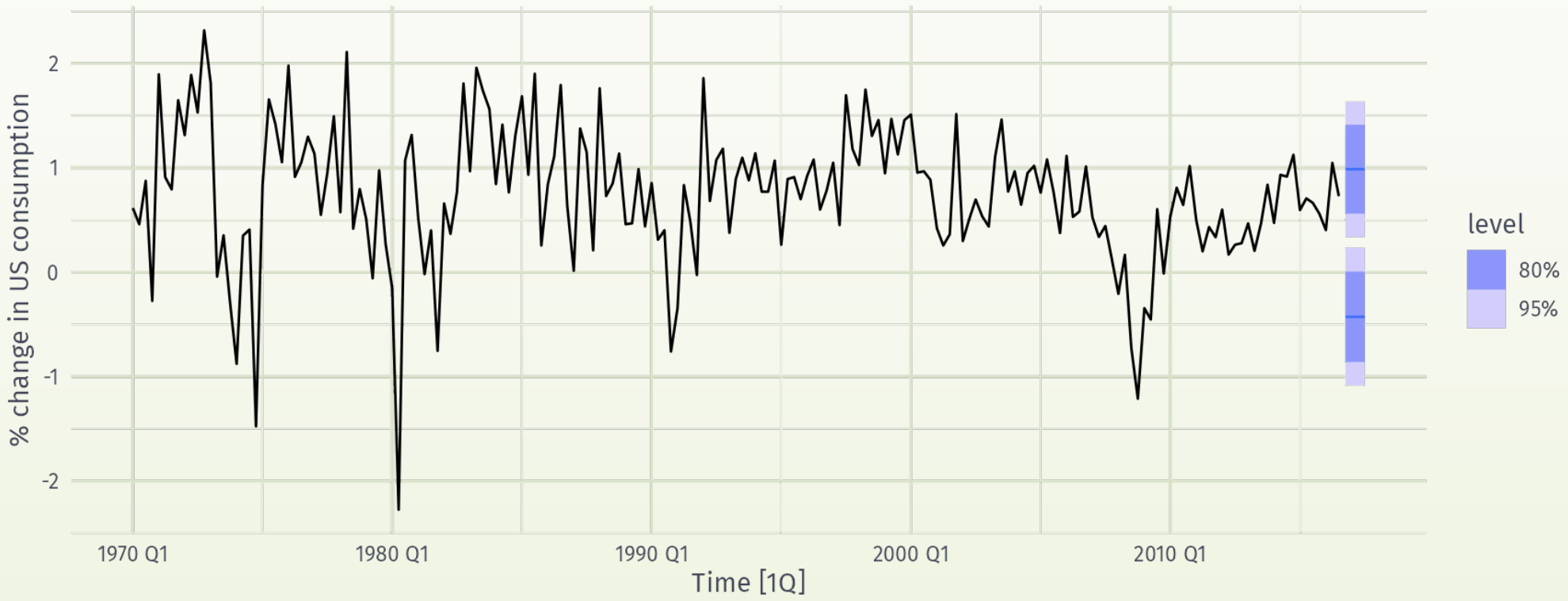
```
fit_consBest <- us_change %>%
  model(
    TSLM(Consumption ~ Income + Savings + Unemployment)
  )

down_future <- new_data(us_change, 4) %>%
  mutate(Income = -1, Savings = -0.5, Unemployment = 0)
fc_down <- forecast(fit_consBest, new_data = down_future)

up_future <- new_data(us_change, 4) %>%
  mutate(Income = 1, Savings = 0.5, Unemployment = 0)
fc_up <- forecast(fit_consBest, new_data = up_future)
```

US Consumption

```
us_change %>% autoplot(Consumption) +
  ylab("% change in US consumption") +
  autolayer(fc_up, series = "increase") +
  autolayer(fc_down, series = "decrease") +
  guides(colour = guide_legend(title = "Scenario"))
```



SECTION 5 OF 6

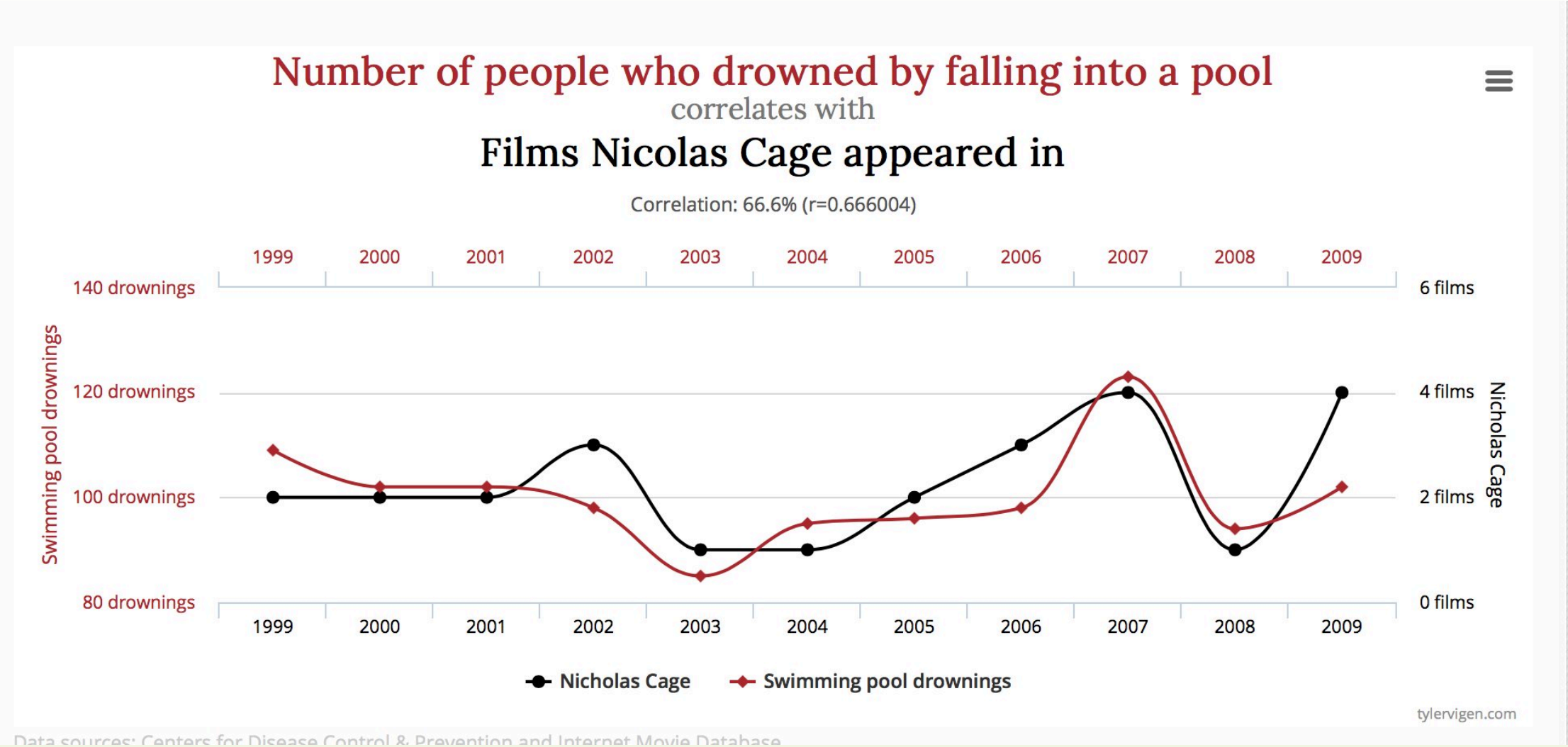
Correlation, causation and forecasting

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Correlation does not imply causation

Check out https://www.tylervigen.com/spurious-correlations



Correlation is not causation

- When x is useful for predicting y, it is not necessarily causing y.
- e.g., predict number of drownings y using number of ice-creams sold x.
- Correlations are useful for forecasting, even when there is no causality.
- Better models usually involve causal relationships (e.g., temperature x and people z to predict drownings y).

Multicollinearity

In regression analysis, multicollinearity occurs when:

- Two predictors are highly correlated (i.e., the correlation between them is close to ± 1).
- A linear combination of some of the predictors is highly correlated with another predictor.
- A linear combination of one subset of predictors is highly correlated with a linear combination of another subset of predictors.

Multicollinearity

If multicollinearity exists...

- the numerical estimates of coefficients may be wrong (worse in Excel than in a statistics package)
- don't rely on the p-values to determine significance.
- there is no problem with model *predictions* provided the predictors used for forecasting are within the range used for fitting.
- omitting variables can help.
- combining variables can help.

Outliers and influential observations

Things to watch for

- *Outliers*: observations that produce large residuals.
- *Influential observations*: removing them would markedly change the coefficients. (Often outliers in the x variable).
- *Lurking variable*: a predictor not included in the regression but which has an important effect on the response.
- Points should not normally be removed without a good explanation of why they are different.

Modern regression models

- Suppose instead of 3 regressors we had 44.
  - For example, 44 predictors leads to 18 trillion possible models!
- Stepwise regression cannot solve this problem due to the number of variables.
- We need to use the family of Lasso models: lasso, ridge, elastic net

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Some useful predictors for regression models

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Dummy variables

If a categorical variable takes only two values (e.g., “Yes” or “No”), then an equivalent numerical variable can be constructed taking value 1 if yes and 0 if no. This is called a **dummy variable**.

	A	B
1	Yes	1
2	Yes	1
3	No	0
4	Yes	1
5	No	0
6	No	0
7	Yes	1
8	Yes	1
9	No	0
10	No	0
11	No	0
12	No	0
13	Yes	1
14	No	0

Dummy variables

If there are more than two categories, then the variable can be coded using several dummy variables (one fewer than the total number of categories).

	A	B	C	D	E
1	Monday	1	0	0	0
2	Tuesday	0	1	0	0
3	Wednesday	0	0	1	0
4	Thursday	0	0	0	1
5	Friday	0	0	0	0
6	Monday	1	0	0	0
7	Tuesday	0	1	0	0
8	Wednesday	0	0	1	0
9	Thursday	0	0	0	1
10	Friday	0	0	0	0
11	Monday	1	0	0	0
12	Tuesday	0	1	0	0
13	Wednesday	0	0	1	0
14	Thursday	0	0	0	1
15	Friday	0	0	0	0

Beware of the dummy variable trap!

- Using one dummy for each category gives too many dummy variables!
- The regression will then be singular and inestimable.
- Either omit the constant, or omit the dummy for one category.
- The coefficients of the dummies are relative to the omitted category.

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies

Outliers

- If there is an outlier, you can use a dummy variable to remove its effect.

Public holidays

- For daily data: if it is a public holiday, dummy=1, otherwise dummy=0.

Intervention variables

Spikes

- Equivalent to a dummy variable for handling an outlier.

Steps

- Variable takes value 0 before the intervention and 1 afterwards.

Change of slope

- Variables take values 0 before the intervention and values $\{1, 2, 3, \ldots\}$ afterwards.
- this could be also handled using `trend()`

Include special event using dummies

- Christmas Eve: if Christmas Eve, $v_t = 1$, $v_t = 0$ otherwise
- New year's Day: if New year's Day, $v_t = 1$, $v_t = 0$ otherwise.
- and more: Ramadan and Chinese new year, school holiday, etc

Interactions

For example, sometimes the effect of a particular event might be different if it is on a weekend or a week day or its effect might be different in each shift:

- you need to introduce an interaction variable
- you can use a new dummy as : `v1*v2`

Lagged predictors

The model include present and past values of predictor: $x_t, x_{t-1}, x_{t-2}, \ldots$.

$$y_t = a + \beta_0 x_t + \beta_1 x_{t-1} + \cdots + \beta_k x_{t-k} + \varepsilon_t$$

- x can influence y, but y is not allowed to influence x.

Lagged predictors

- Lagged values of a predictor:
  - Create new variables by shifting the existing variable backwards

Example: x is advertising which has a delayed effect

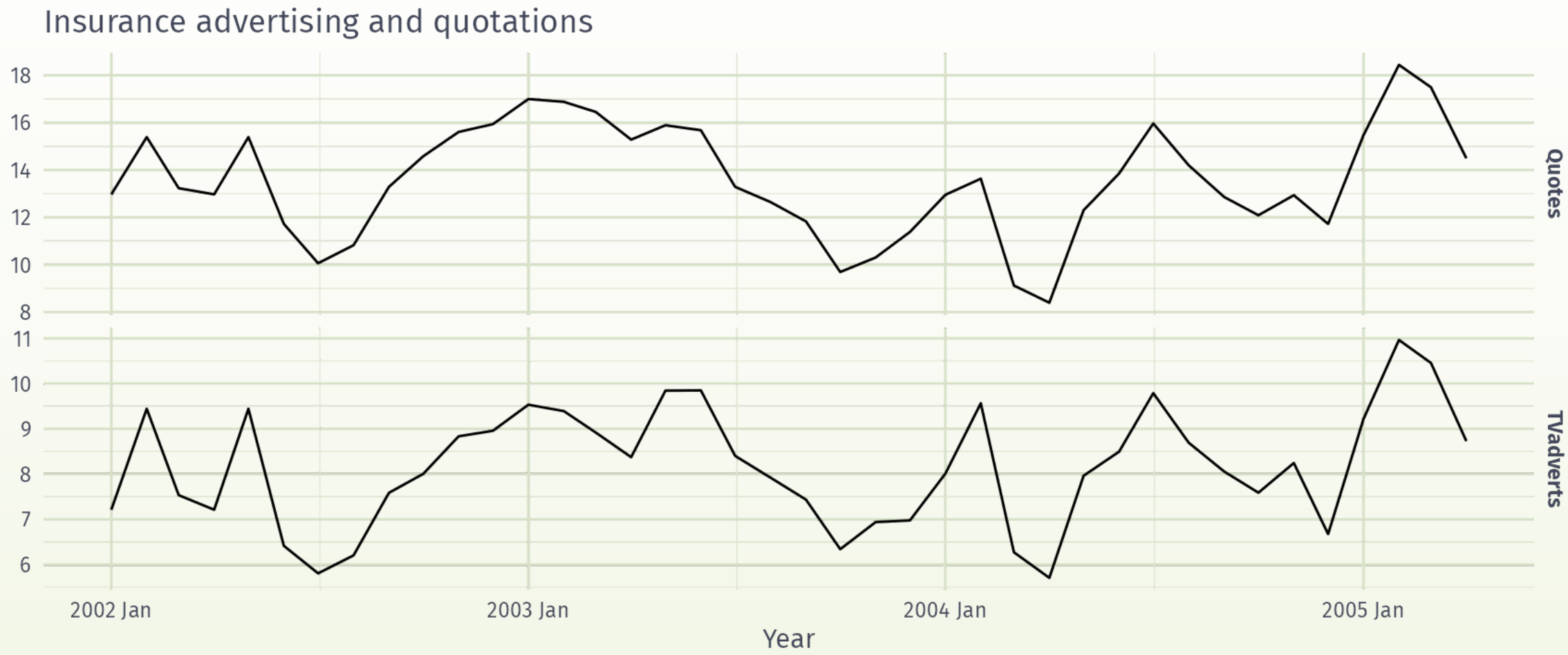
$$
\begin{aligned}
x_1 &= \text{advertising for previous month;} \\
x_2 &= \text{advertising for two months previously;} \\
&\vdots \\
x_m &= \text{advertising for } m \text{ months previously.}
\end{aligned}
$$

Example: Insurance quotes and TV adverts

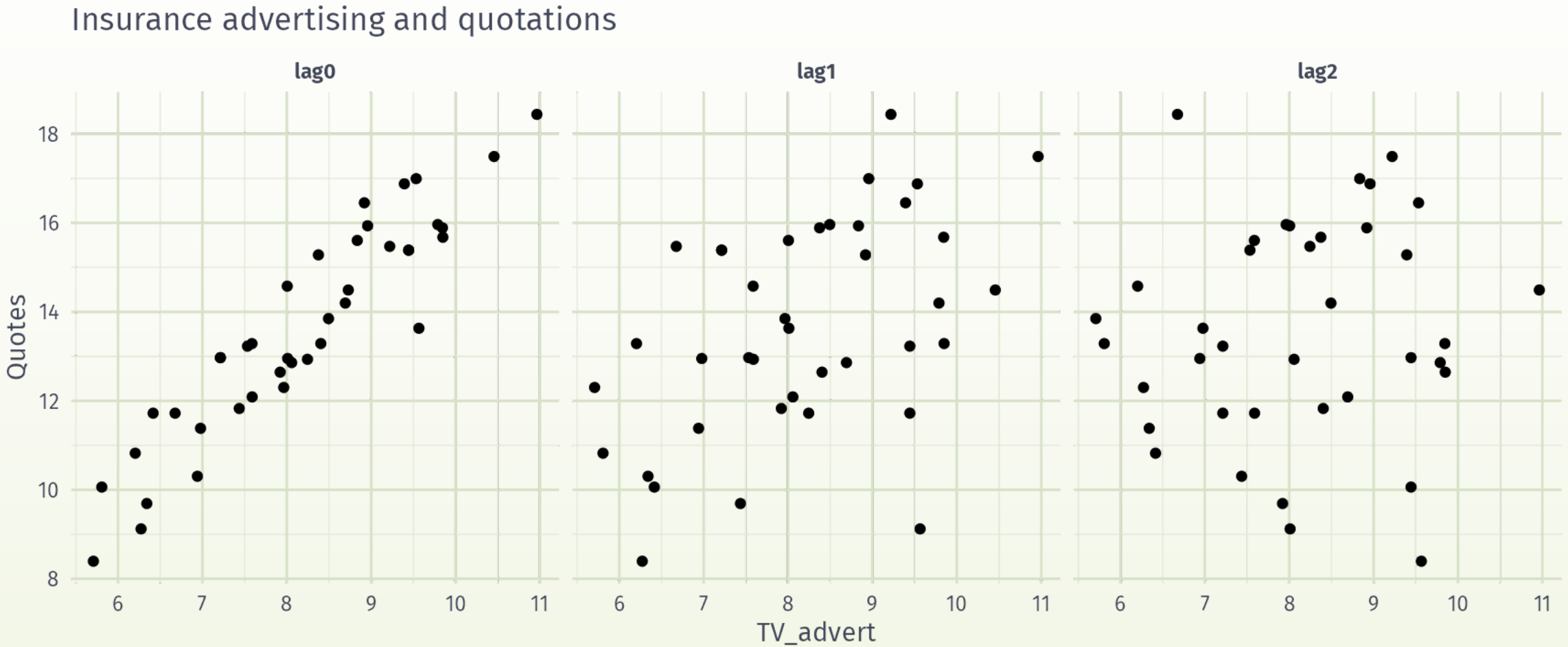
```
insurance
```

```
# A tsibble: 40 x 3 [1M]
      Month Quotes TVadverts
      <mth>  <dbl>     <dbl>
 1 2002 Jan   13.0      7.21
 2 2002 Feb   15.4      9.44
 3 2002 Mar   13.2      7.53
 4 2002 Apr   13.0      7.21
 5 2002 May   15.4      9.44
 6 2002 Jun   11.7      6.42
 7 2002 Jul   10.1      5.81
 8 2002 Aug   10.8      6.20
 9 2002 Sep   13.3      7.59
10 2002 Oct   14.6      8.00
# i 30 more rows
```

Example: Insurance quotes and TV adverts



Example: Insurance quotes and TV adverts



Example: Insurance quotes and TV adverts

```
fit <- insurance |>
  # Restrict data so models use same fitting period
  mutate(Quotes = c(NA, NA, NA, Quotes[4:40])) |>
  model(
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts),
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts +
      lag(TVadverts)),
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts +
      lag(TVadverts) +
      lag(TVadverts, 2)),
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts +
      lag(TVadverts) +
      lag(TVadverts, 2) +
      lag(TVadverts, 3))
  )
```

Example: Insurance quotes and TV adverts

```
glance(fit)
```

Lag order	sigma2	log_lik	AIC	AICc	BIC
0	0.265	-28.3	66.6	68.3	75.0
1	0.209	-24.0	58.1	59.9	66.5
2	0.215	-24.0	60.0	62.6	70.2
3	0.206	-22.2	60.3	65.0	73.8

Example: Insurance quotes and TV adverts

```
# Re-fit to all data
fit <- insurance |>
  model(ARIMA(Quotes ~ TVadverts + lag(TVadverts) + pdq(d = 0)))
report(fit)
```

```
Series: Quotes
Model: LM w/ ARIMA(1,0,2) errors

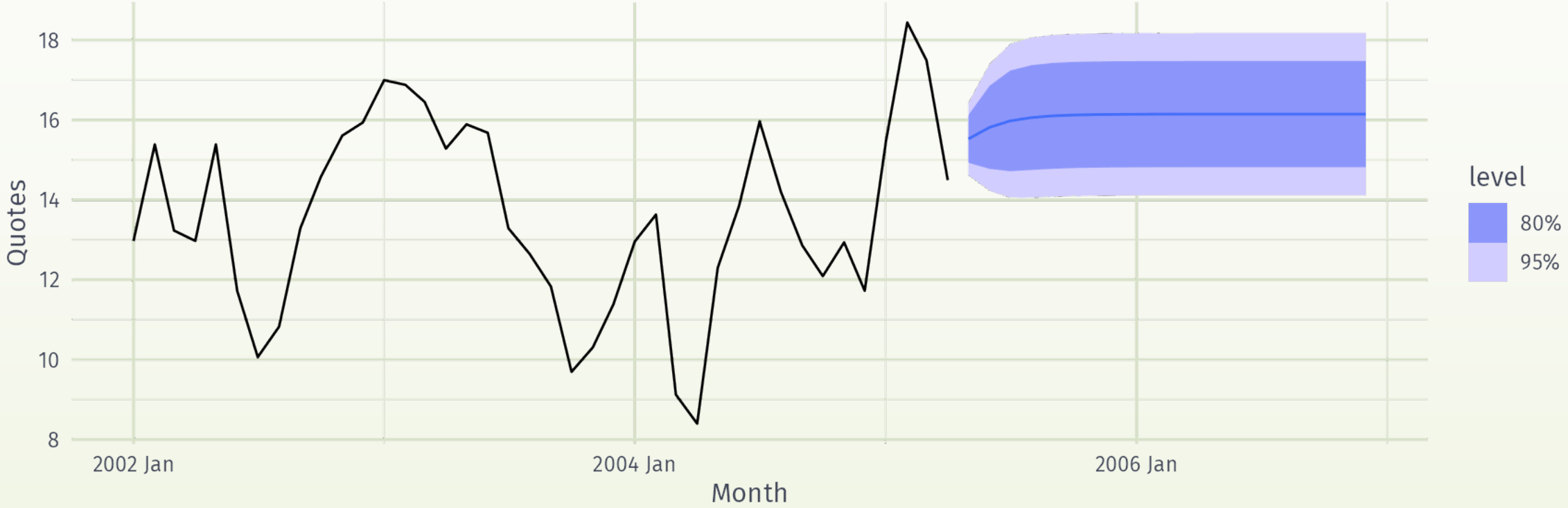
Coefficients:
        ar1    ma1    ma2  TVadverts  lag(TVadverts)  intercept
      0.512  0.917  0.459     1.2527          0.1464       2.16
s.e.  0.185  0.205  0.190     0.0588          0.0531       0.86

sigma^2 estimated as 0.2166:  log likelihood=-23.9
AIC=61.9   AICc=65.4   BIC=73.7
```

$$y_t = 2.16 + 1.25x_t + 0.15x_{t-1} + \eta_t,$$
$$\eta_t = 0.512\eta_{t-1} + \varepsilon_t + 0.92\varepsilon_{t-1} + 0.46\varepsilon_{t-2}.$$

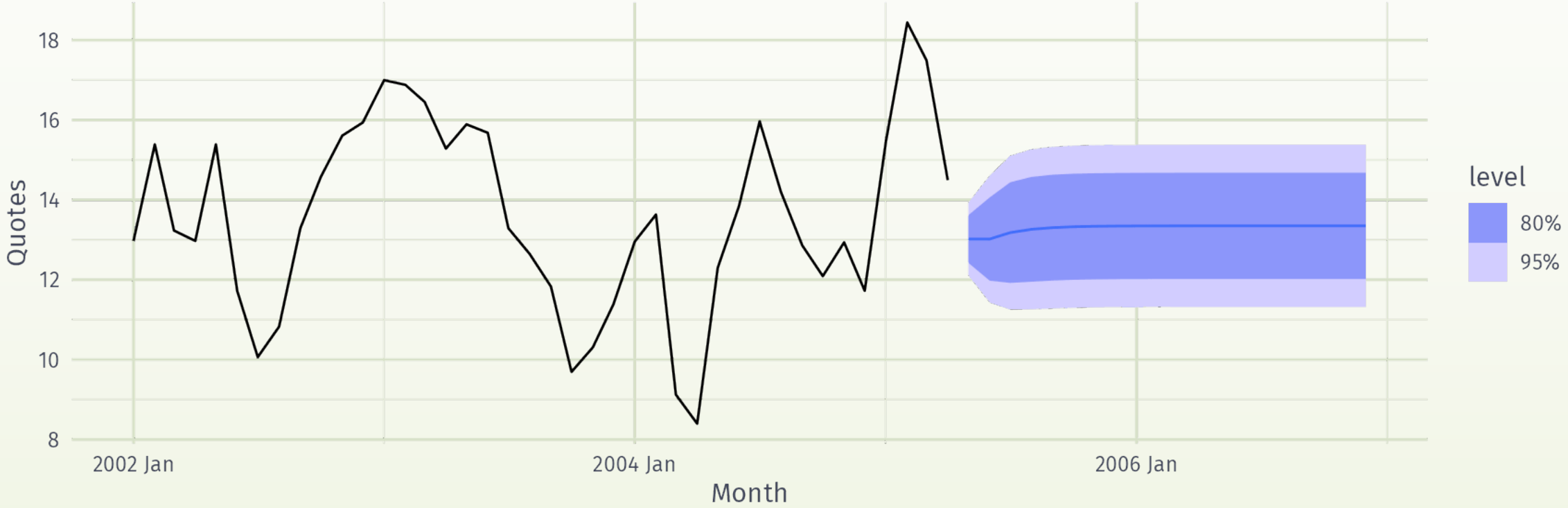
Example: Insurance quotes and TV adverts

```
advert_a <- new_data(insurance, 20) |>
  mutate(TVadverts = 10)
forecast(fit, advert_a) |> autoplot(insurance)
```



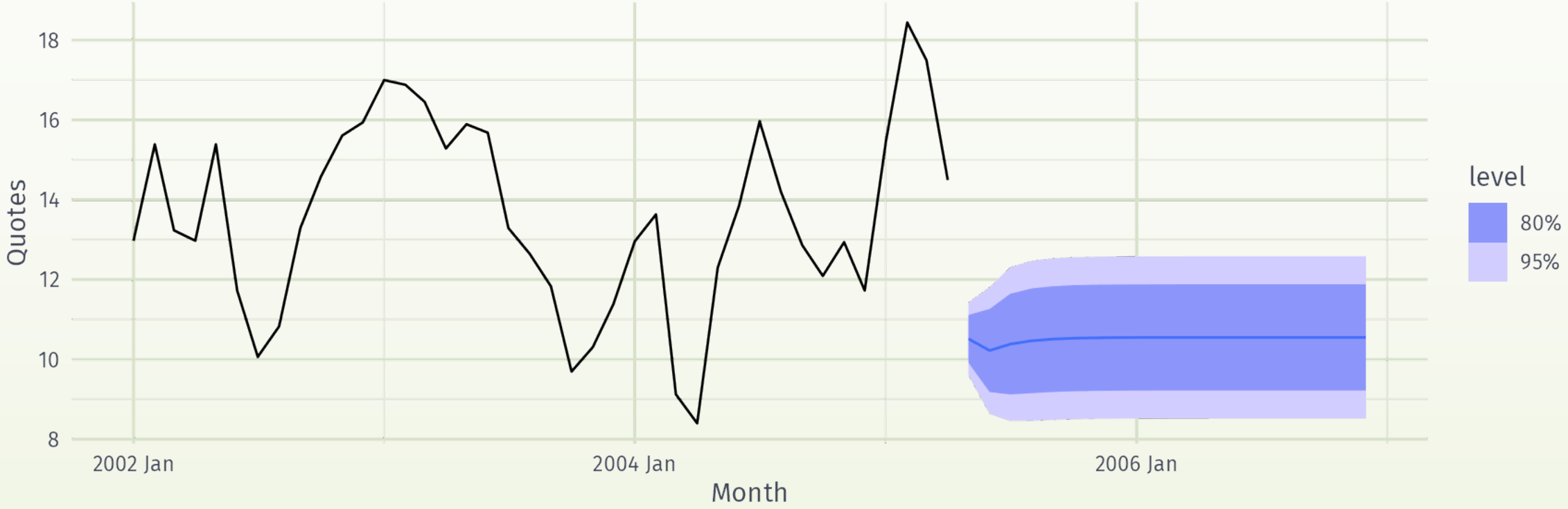
Example: Insurance quotes and TV adverts

```
advert_b <- new_data(insurance, 20) |>
  mutate(TVadverts = 8)
forecast(fit, advert_b) |> autoplot(insurance)
```



Example: Insurance quotes and TV adverts

```
advert_c <- new_data(insurance, 20) |>
  mutate(TVadverts = 6)
forecast(fit, advert_c) |> autoplot(insurance)
```



Lead predictors

Sometimes a change in the predictor x_t that will happen in the future will affect the value of y_t in the past. We say x_t is a leading indicator.

- Lead values of a predictor:
  - Create new variables by shifting the existing variable forwards

- $y_t =$ sales, $x_t =$ tax policy announcement